

Slide 3: Comparing Strings

*** Unit 1 Lecture***

Slide 1: Intro to Python and Environment Setup

```
print("We can display aka print Strings in python using print function.")
```

We can display aka print Strings in python using print function.

```
print("""
We can display a whole block of text by using a text block
Introduction to Google Colab:
Google Colab is a free, cloud-based Jupyter notebook environment provided by Google. It supports Python and is ideal

## Setting Up Your Environment
1. Sign in to your Google account.
2. Open [Google Colab](https://colab.research.google.com).
3. Save a copy of this notebook to your drive.
""")
```

```
We can display a whole block of text by using a text block
Introduction to Google Colab:
Google Colab is a free, cloud-based Jupyter notebook environment provided by Google. It supports Python and is ideal

## Setting Up Your Environment
1. Sign in to your Google account.
2. Open [Google Colab](https://colab.research.google.com).
3. Save a copy of this notebook to your drive.
```

```
print('This will also work.')
```

This will also work.

Slides 2- 3: Displaying Output with the print Function

```
# Displaying output with the print function
print('Hello, world!')
```

Hello, world!

Challenge 1

```
#Vikram Nadathur May 26th 2026
# Challenge: Print a greeting with your name

# Add a comment that the following is a greeting to your faculty
#And Print a greeting to your faculty
# This is a greeting to my faculty
print('Hello, Vikram!')
print('Hello, Professor!')
```

Hello, Vikram!
Hello, Professor!

Slide 4-6: Variables, Reading Input, and Performing Calculations

```
# Variables and Reading Input
name = input('What is your name? ')
```

```
food = input('Enter your favorite food: ')
year = int(input('Enter current year: '))
print(f'Hello, {name}! Your favorite food in {year} is {food}.')
```

```
What is your name? Vikram
Enter your favorite food: Pasta
Enter current year: 2026
Hello, Vikram! Your favorite food in 2026 is Pasta.
```

▼ Challenge 2

```
#Vikram Nadathur May 26th 2026
# Challenge: Create a program that asks for two numbers and prints their sum, difference, product, and quotient.
num1 = float(input('Enter the first number: '))
num2 = float(input('Enter the second number: '))

sum_result = num1 + num2
diff_result = num1 - num2
prod_result = num1 * num2

print(f'Sum: {sum_result}')
print(f'Difference: {diff_result}')
print(f'Product: {prod_result}')

if num2 != 0:
    quot_result = num1 / num2
    print(f'Quotient: {quot_result}')
else:
    print('Quotient: undefined (cannot divide by zero)')

#... complete the task
#... print all results
```

```
Enter the first number: 5
Enter the second number: 7
Sum: 12.0
Difference: -2.0
Product: 35.0
Quotient: 0.7142857142857143
```

▼ Slide 7: String Concatenation

```
# String concatenation
greeting = 'Hello'
name = 'Alice'
message = greeting + ', ' + name + '!'
print(message)
```

```
Hello, Alice!
```

▼ Challenge 3

```
#Vikram Nadathur May 26th 2026
# Challenge: Create a personalized message by concatenating strings
#obtain first name, last name and other information from the user and print for them the summary of collected inform
first_name = input('Enter your first name: ')
last_name = input('Enter your last name: ')
age = input('Enter your age: ')
hobby = input('Enter your favorite hobby: ')

personal_message = (
    'Hello, ' + first_name + ' ' + last_name + '!'
    + 'You are ' + age + ' years old and you enjoy ' + hobby + '!'
)

print(personal_message)
```

```
Enter your first name: Vikram
Enter your last name: Nadathur
Enter your age: 26
```

```
Enter your favorite hobby: swimming
Hello, Vikram Nadathur! You are 26 years old and you enjoy swimming.
```

Slide 8-10: More About the print Function

```
print('Hello', 'world', sep='-', end='!')

Hello-world!
```

Slide 11-12: Displaying Formatted Output

```
name = 'Alice'
age = 25
print(f'{name} is {age} years old.')

product = 'Laptop'
price = 999.99
print(f'The price of the {product} is ${price:.2f}.')

Alice is 25 years old.
The price of the Laptop is $999.99.
```

Challenge 4

```
#Vikram Nadathur May 26th 2026
## Print a customer receipt with 5 different products and very different prices as a table
# 1. Store products and prices in a list of tuples
products = [
    ('Pencil', 0.99),
    ('Notebook', 2.49),
    ('Backpack', 29.99),
    ('Calculator', 15.5),
    ('USB Drive', 8.75)
]

# 2. Print the header
print('Customer Receipt')
print('-' * 30)
print(f'{"Item":15}{"Price":>15}')
print('-' * 30)

# 3. Loop through the list and print each item dynamically
for item, price in products:
    print(f'{item:15}${price:>14.2f}')

print('-' * 30)

# 4. Use sum() to calculate the total automatically
total = sum(price for item, price in products)
print(f'{"Total":15}${total:>14.2f}')
```

Customer Receipt

Item	Price
Pencil	\$ 0.99
Notebook	\$ 2.49
Backpack	\$ 29.99
Calculator	\$ 15.50
USB Drive	\$ 8.75
Total	\$ 57.72

Slide 13: Named Constants

```
# Using named constants
INTEREST_RATE = 0.05
```

```
principal = 1000
interest = principal * INTEREST_RATE
print(f'The interest is {interest:.2f}.')
```

The interest is 50.00.

▼ Challenge 5

```
#Vikram Nadathur May 26th 2026
# Challenge: Define and use named constants in a simple calculation
TAX_RATE = 0.07
price = 50
tax = price * TAX_RATE
print(f'The tax on an item costing ${price} is ${tax:.2f}.')
```

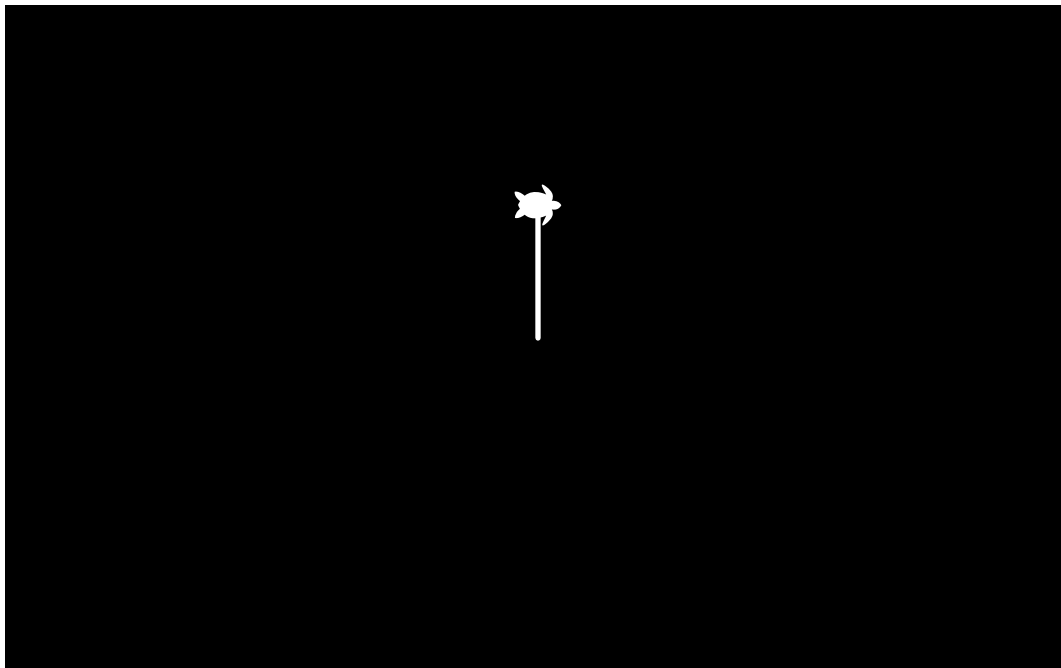
The tax on an item costing \$50 is \$3.50.

▼ Slide 14-15: Introduction to Turtle Graphics

```
!pip install --upgrade ColabTurtle
```

```
Collecting ColabTurtle
  Downloading ColabTurtle-2.1.0.tar.gz (6.8 kB)
  Preparing metadata (setup.py) ... done
Building wheels for collected packages: ColabTurtle
  Building wheel for ColabTurtle (setup.py) ... done
  Created wheel for ColabTurtle: filename=ColabTurtle-2.1.0-py3-none-any.whl size=7643 sha256=f1cff85c78e258b0474e3e9
  Stored in directory: /root/.cache/pip/wheels/9f/af/64/ffd85f9858ed7d56b7293dcedbc9d461bf13c8cfc97e352bc8
Successfully built ColabTurtle
Installing collected packages: ColabTurtle
Successfully installed ColabTurtle-2.1.0
```

```
# Introduction to turtle graphics
from ColabTurtle.Turtle import *
initializeTurtle()
forward(100)
right(90)
speed(8)
```

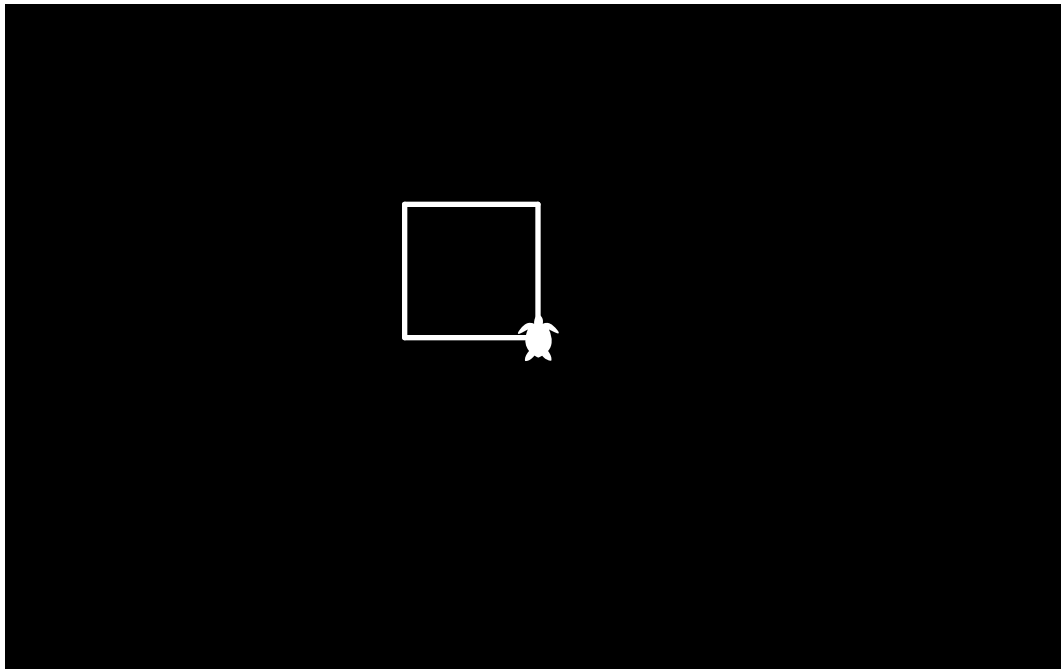


▼ Challenge 6

```
#Vikram Nadathur May 26th 2026
# Challenge: Draw a square using the turtle module

from ColabTurtle.Turtle import *

initializeTurtle()
for _ in range(4):
    forward(100)
    left(90)
```



Slide 16: Introduction to File Input and Output

Mount drive to be able to store files on your google drive

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

create a folder Unit and example.txt in it

```
import os

directory_path = '/content/drive/MyDrive/Unit1/'
os.makedirs(directory_path, exist_ok=True)

with open(os.path.join(directory_path, 'example.txt'), 'w') as file:
    file.write('Hello, world!')

with open(os.path.join(directory_path, 'example.txt'), 'r') as file:
    content = file.read()
    print(content)
```

Hello, world!

Challenge 7

```
#Vikram Nadathur May 26th 2026
# Challenge: Create a program that writes a list of numbers to a file and then reads them back
numbers = [1, 2, 3, 4, 5]
```

```
with open('numbers.txt', 'w') as file:
    for number in numbers:
        file.write(f'{number}\\n')

with open('numbers.txt', 'r') as file:
    content = file.read()
    print(content)
```

```
1\n2\n3\n4\n5\n
```

Slide 17: Using Loops to Process Files

```
# Using loops to process files
with open('numbers.txt', 'r') as file:
    for line in file:
        print(line.strip())
```

```
1\n2\n3\n4\n5\n
```

Challenge 8

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that reads a file and prints only the even numbers
with open('numbers.txt', 'r') as file:
    content = file.read() # Read the whole content as one string
    numbers_as_strings = content.split('\\n') # Split by the literal '\n' string

    for num_str in numbers_as_strings:
        cleaned_num_str = num_str.strip()
        if cleaned_num_str: # Ensure it's not an empty string
            number = int(cleaned_num_str)
            if number % 2 == 0:
                print(number)
```

```
2
4
```

Slide 18: Using the with Statement to Open Files

```
# Using the with statement to open files
# The file was created in '/content/drive/MyDrive/Unit1/' in a previous cell.
# We need to specify the full path to access it.
import os

directory_path = '/content/drive/MyDrive/Unit1/'
file_path = os.path.join(directory_path, 'example.txt')

with open(file_path, 'r') as file:
    content = file.read()
    print(content)
```

```
Hello, world!
```

Challenge 9

```
#Vikram Nadathur May 26th 2026
# Challenge: Use the with statement to write and read multiple lines
lines = ['Line 1', 'Line 2', 'Line 3']
with open('multilines.txt', 'w') as file:
    for line in lines:
        file.write(line + '\\n')

with open('multilines.txt', 'r') as file:
    for line in file:
        print(line.strip())
```

```
Line 1\nLine 2\nLine 3\n
```

Slide 19: Processing Records

```
# Processing records
records = [
    {'name': 'Alice', 'age': 25},
    {'name': 'Bob', 'age': 30}
]
for record in records:
    print(f"{record['name']} is {record['age']} years old.")
```

Alice is 25 years old.
Bob is 30 years old.

Challenge 10

```
#Vikram Nadathur May 26th 2026
# Challenge: Add a new record and print all records
records.append({'name': 'Charlie', 'age': 35})
for record in records:
    print(f"{record['name']} is {record['age']} years old.")
```

Alice is 25 years old.
Bob is 30 years old.
Charlie is 35 years old.

Slide 20: Exceptions

```
# Handling exceptions
try:
    x = 10 / 0
except ZeroDivisionError:
    print('Cannot divide by zero!')
```

Cannot divide by zero!

```
# Handle multiple exceptions
try:
    x = int(input('Enter a number: '))
    y = int(input('Enter another number: '))
    result = x / y
    print(f'The result is {result}')
except ValueError:
    print('Invalid input!')
except ZeroDivisionError:
    print('Cannot divide by zero!')
```

Enter a number: 70
Enter another number: 0
Cannot divide by zero!

*** Unit 2 Lecture***

Slide 1: The if Statement

```
# The if statement
x = 10
if x > 5:
    print("x is greater than 5")
```

x is greater than 5

Challenge 11

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that checks if a number is positive, negative, or zero.
num = int(input("Enter a number: "))

if num > 0:
    print("The number is positive")
elif num < 0:
    print("The number is negative")
else:
    print("The number is zero")

# COMPLETE THIS CODE
```

```
Enter a number: 70
The number is positive
```

Slide 2: The if-else Statement

```
# The if-else statement
x = 10
if x > 15:
    print("x is greater than 15")
else:
    print("x is not greater than 15")
```

```
x is not greater than 15
```

Challenge 12

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that determines if a person is eligible to vote based on age.
age = int(input("Enter your age: "))

if age >= 18:
    print("You are eligible to vote.")
else:
    print("You are not eligible to vote yet.")
# COMPLETE THIS CODE
```

```
Enter your age: 26
You are eligible to vote.
```

Slide 3: Comparing Strings

```
# Comparing strings
string1 = "hello"
string2 = "world"
if string1 == string2:
    print("The strings are equal")
else:
    print("The strings are not equal")
```

```
The strings are not equal
```

Challenge 13


```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that checks if a string is a palindrome.
string = input("Enter a string: ")

# Remove spaces and make case-insensitive
cleaned = string.replace(' ', '').lower()

if cleaned == cleaned[::-1]:
    print("The string is a palindrome.")
else:
    print("The string is not a palindrome.")
# COMPLETE THIS CODE
```

```
Enter a string: racecar
The string is a palindrome.
```

Slide 4: Nested Decision Structures and the if-elif-else Statement

```
score = int(input("Enter your score: "))
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
elif score >= 60:
    print("Grade: D")
else:
    print("Grade: F")
```

```
Enter your score: 90
Grade: A
```

Challenge 14

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that calculates the discount for a purchase based on the amount.
amount = float(input("Enter the purchase amount: "))

if amount >= 100:
    discount_rate = 0.20
elif amount >= 50:
    discount_rate = 0.10
else:
    discount_rate = 0.0

discount = amount * discount_rate
final_price = amount - discount

print(f'Purchase amount: ${amount:.2f}')
print(f'Discount: ${discount:.2f}')
print(f'Final price: ${final_price:.2f}')
# COMPLETE THIS CODE
```

```
Enter the purchase amount: 100.00
Purchase amount: $100.00
Discount: $20.00
Final price: $80.00
```

Slide 5: Logical Operators

```
# Logical operators
x = 10
y = 20
if x > 5 and y > 15:
    print("Both conditions are true")
if x > 15 or y > 15:
    print("At least one condition is true")
if not (x > 15):
    print("x is not greater than 15")
```

```
Both conditions are true
At least one condition is true
x is not greater than 15
```

▼ Challenge 15

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that checks if a number is within a specific range.
num = int(input("Enter a number: "))

lower_bound = 10
upper_bound = 20

if lower_bound <= num <= upper_bound:
    print(f'The number is within the range {lower_bound} to {upper_bound}.')
else:
    print(f'The number is outside the range {lower_bound} to {upper_bound}.')

# COMPLETE THIS CODE
```

```
Enter a number: 4
The number is outside the range 10 to 20.
```

▼ Slide 6: Boolean Variables

```
# Boolean variables
is_raining = True
if is_raining:
    print("It is raining")
else:
    print("It is not raining")
```

```
It is raining
```

▼ Challenge 16

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that uses a boolean variable to check if a number is even or odd.
num = int(input("Enter a number: "))

is_even = (num % 2 == 0)

if is_even:
    print("The number is even.")
else:
    print("The number is odd.")
# COMPLETE THIS CODE
```

```
Enter a number: 14
The number is even.
```

▼ Slide 7: Conditional Expressions

```
# Conditional expressions
score = 75
result = "Pass" if score >= 60 else "Fail"
print(result)
```

```
Pass
```

▼ Challenge 17

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that assigns a grade based on the score using a conditional expression.
score = int(input("Enter your score: "))
```

```

grade = (
    "A" if score >= 90 else
    "B" if score >= 80 else
    "C" if score >= 70 else
    "D" if score >= 60 else
    "F"
)

print(f'Your grade is: {grade}')
# COMPLETE THIS CODE

```

```

Enter your score: 85
Your grade is: B

```

Slide 8: Assignment Expressions and the Walrus Operator

```

# Assignment expressions and the walrus operator
print(num := 99)
if (area := 20 * 5) > 100:
    print("The area is too large")

```

```
99
```

Challenge 18

```

#Vikram Nadathur May 26th 2026
# Challenge: Write a program that uses the walrus operator to read input and validate it.
while (num := int(input("Enter a positive number: "))) <= 0:
    print("The number must be positive")
print(f"You entered: {num}")

```

```

Enter a positive number: 4
You entered: 4

```

Slide 9: Introduction to Repetition Structures

Double-click (or enter) to edit

```

# Introduction to repetition structures
for i in range(5):
    print(i)

```

```

0
1
2
3
4

```

Challenge 19

```

#Vikram Nadathur May 26th 2026
# Challenge: Write a program that prints the first 10 even numbers using a loop.
for num in range(11):
    if num%2==0:
        print(num)
# COMPLETE THIS CODE

```

```

0
2
4
6
8
10

```

Slide 10: Loops

```
# The while loop
n = 0
while n < 5:
    print(n)
    n += 1
```

```
0
1
2
3
4
```

Challenge 20

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that keeps asking the user for a number until they enter 0.
while True:
    num = int(input("Enter a number (enter 0 to quit): "))
    if num == 0:
        break
    print(f"You entered: {num}")
# COMPLETE THIS CODE
```

```
Enter a number (enter 0 to quit): 8
You entered: 8
Enter a number (enter 0 to quit): 9
You entered: 9
Enter a number (enter 0 to quit): 3
You entered: 3
Enter a number (enter 0 to quit): 4
You entered: 4
Enter a number (enter 0 to quit): 55
You entered: 55
Enter a number (enter 0 to quit): 0
```

```
# The for loop
for i in range(5):
    print(i)
```

```
0
1
2
3
4
```

Challenge 21

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that prints the squares of numbers from 1 to 10.
for num in range(11):
    print(num*num)
# COMPLETE THIS CODE
```

```
0
1
4
9
16
25
36
49
64
81
100
```

```
# Calculating a running total
total = 0
for i in range(5):
    num = int(input("Enter a number: "))
```

```
total += num
print(f"Total: {total}")
```

```
Enter a number: 8
Enter a number: 9
Enter a number: 9
Enter a number: -9
Enter a number: 7
Total: 24
```

Challenge 22

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that calculates the sum of numbers entered by the user until they enter a negative numb
total = 0
while True:
    num = int(input("Enter a number (enter a negative number to quit): "))
    if num < 0:
        break
    total += num
print(f"Total: {total}")
# COMPLETE THIS CODE
```

```
Enter a number (enter a negative number to quit): 4
Enter a number (enter a negative number to quit): 3
Enter a number (enter a negative number to quit): 2
Enter a number (enter a negative number to quit): 1
Enter a number (enter a negative number to quit): 0
Enter a number (enter a negative number to quit): -1
Total: 10
```

Slide 13: Sentinels

```
# Using a sentinel value
total = 0
num = int(input("Enter a number (-1 to quit): "))
while num != -1:
    total += num
    num = int(input("Enter a number (-1 to quit): "))
print(f"Total: {total}")
```

```
Enter a number (-1 to quit): 2
Enter a number (-1 to quit): -1
Total: 2
```

Challenge 23

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that calculates the average of numbers entered by the user using -1 as a sentinel value
count=0
while True:
    num = int(input("Enter a number (enter -1 to quit): "))
    if num == -1:
        break
    total += num
    count += 1
print(f"Average: {total/count}")

# COMPLETE THIS CODE
```

```
Enter a number (enter -1 to quit): 4
Enter a number (enter -1 to quit): 3
Enter a number (enter -1 to quit): 2
Enter a number (enter -1 to quit): 1
Enter a number (enter -1 to quit): -1
Average: 5.25
```

```
# Input validation loops
num = int(input("Enter a positive number: "))
while num <= 0:
    print("The number must be positive")
```

```
num = int(input("Enter a positive number: "))
print(f"You entered: {num}")
```

```
Enter a positive number: 4
You entered: 4
```

Challenge 24

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that asks for a password and keeps asking until the correct password is entered.
correct_password = "python123"
password = input("Enter the password: ")

while password != correct_password:
    print("Incorrect password, try again.")
    password = input("Enter the password: ")

print("Access granted.")
# COMPLETE THIS CODE
```

```
Enter the password: john
Incorrect password, try again.
Enter the password: python123
Access granted.
```

```
# Nested loops
for i in range(3):
    for j in range(3):
        print(f"i: {i}, j: {j}")
```

```
i: 0, j: 0
i: 0, j: 1
i: 0, j: 2
i: 1, j: 0
i: 1, j: 1
i: 1, j: 2
i: 2, j: 0
i: 2, j: 1
i: 2, j: 2
```

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that prints a multiplication table from 1 to 5 using nested loops.
# COMPLETE THIS CODE
for i in range(1, 6):
    for j in range(1, 6):
        print(f'{i} x {j} = {i * j}')
    print('-' * 15)
```

```
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
1 x 5 = 5
-----
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
-----
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
-----
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
-----
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
```

$$5 \times 5 = 25$$

- Slide 12: The break Statement

```
# The break statement
n = 0
while n < 10:
    if n == 5:
        break
    print(n)
    n += 1
```

0
1
2
3
4

▼ Challenge 25

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that stops asking for input when the user enters 'stop'.
# COMPLETE THIS CODE
while True:
    user_input = input("Enter something (type 'stop' to end): ")
    if user_input.lower() == 'stop':
        break
    print(f'You entered: {user_input}')
```

```
Enter something (type 'stop' to end): hello
You entered: hello
Enter something (type 'stop' to end): world
You entered: world
Enter something (type 'stop' to end): python
You entered: python
Enter something (type 'stop' to end): is cool
You entered: is cool
Enter something (type 'stop' to end): stop
```

- Slide 13: The continue Statement

```
# The continue statement
for i in range(10):
    if i % 2 == 0:
        continue
    print(i)
```

1
3
5
7
9

▼ Challenge 26

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that prints all numbers from 1 to 10 except 5.
# COMPLETE THIS CODE
for i in range(1, 11):
    if i == 5:
        continue
    print(i)
```

- 1
- 2
- 3

```
4
6
7
8
9
10
```

Slide 15: The else Clause with a Loop

```
# The else clause with a loop
for i in range(5):
    print(i)
else:
    print("Loop completed without break")
```

```
0
1
2
3
4
Loop completed without break
```

Challenge 27

```
#Vikram Nadathur May 26th 2026
# Challenge: Write a program that searches for a number in a list and uses the else clause to indicate if it was not
# COMPLETE THIS CODE
numbers = [3, 7, 10, 15, 20]
target = int(input("Enter a number to search for: "))

for n in numbers:
    if n == target:
        print(f'{target} was found in the list.')
        break
else:
    print(f'{target} was not found in the list.')
```

```
Enter a number to search for: 8
8 was not found in the list.
```